

EDUCATION

Rutgers University—New Brunswick | B.S. Electrical & Computer Engineering Expected May 2028
Minor in Computer Science · GPA 3.84/4.00 · Dean's List, all four completed semesters · IEEE–HKN (Eta Kappa Nu) member
Coursework: Data Structures, Digital Logic Design; in progress: Computer Architecture & Assembly, Linear Systems & Signals
Awards: 2nd Place Best Full Paper, IEEE ISEC 2026 · CS Departmental Award, Bergen County Academies (1 of 3 recipients)

RESEARCH

Kuiper TTS: equation-aware speech for scientific papers | Undergraduate Researcher & Co-author 2025–Present
Python · PyTorch Lightning · Piper/VITS · ONNX · LaTeX · Rutgers, 18-member interdisciplinary team

- Co-authored “Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language” (2nd Place Best Full Paper, IEEE ISEC 2026); led the team’s manuscript preparation and reproducibility work.
- Built a reproducible GPU training and ONNX export pipeline for Piper/VITS voices, with a synthesis wrapper, dataset preparation, and a testbench runner; fixed dependency conflicts and validated inference across platforms.
- Wrote Python tools that turn LaTeX into speech-ready narration (fractions, derivatives, nested grouping, operator-aware pacing), with manifests tracing each clip to its source text and settings.

PROJECTS

Room Notes: cross-device notebook | Solo Developer · Private beta 2026–Present
Flutter · Dart · Drift/SQLite · Firebase

- Built one 900 × 1200 page model shared by text, ink, equations, code, and diagrams across desktop, iPad, and phone layouts.
- Designed **local-first sync**: each edit and its pending operation commit in one SQLite transaction, a durable outbox survives restarts, and append-only cloud operations preserve conflicting revisions instead of overwriting them.
- Wrote regression tests for save-and-reopen, interrupted sync, page geometry, and cloud access rules.

Knowsy Breadboard Lab: circuit simulator | Technical Lead, 3-person team 2025–Present
JavaScript · modified nodal analysis · digital simulation · automated testing · Rutgers, with Prof. Kruger

- Led the breadboard module: placed parts and wires become a connectivity model and an inspectable netlist.
- Built an analog solver using **modified nodal analysis** over complex admittances, with frequency sweeps and Bode plots.
- Modeled four-state digital logic (HIGH, LOW, FLOAT, CONFLICT) for 74LS-series parts, so undriven nets and contention are reported, not hidden; verified results against analytical solutions for dividers, filters, and frequency response.

Room of Days: planning & reflection app | Solo Developer · Active development 2025–Present

- Offline-first Flutter app for Android, iOS, and web with validated persistence, backups, cloud sync, and exactly-once rewards.

Smart Plant Care Robot | Embedded Systems Project 2024

- ESP32 irrigation prototype with soil-moisture, light, and environmental sensors; drove pumps through flyback-protected transistor circuits and validated 5–12 V power with multimeter and serial diagnostics.

EXPERIENCE

Schieber Research Group | Software Developer 2023–2024

- Built and documented a full-stack research prototype in React, Vite, and Appwrite, turning study requirements into reusable interfaces and a maintainable web application.

Starbucks | Barista 2023–2025

SKILLS

Languages	Python, C/C++, Java, JavaScript, Dart, SQL, HTML/CSS
AI & research	PyTorch Lightning, Piper/VITS, ONNX, GPU training, LaTeX parsing, experiment reproducibility
Software	Flutter, Firebase, Drift/SQLite, React, Git, Linux, automated testing, local-first sync
Simulation & hardware	Circuit simulation (MNA, AC analysis), digital logic, LTspice, Arduino/ESP32, breadboarding, soldering
Spoken languages	English, Hebrew (fluent); Spanish (conversational)